

**Cost-Aware Routing of Large Language Models: Predicting the Cheapest
Capable Model for Each Request**

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I will build and compare machine learning models that decide which large language model (LLM) a request should be sent to, picking the cheapest model that still answers the request well. The idea is a cost-aware routing layer, an accountant for token spend that watches what each request would cost across models and sends it to the best value option. I will use the LLMRouterBench dataset, which records how a set of LLMs answered the same prompts along with the tokens used, the dollar cost, and a quality score for each answer.

Problem and Proposed Approach

Sending every request to the strongest available model is expensive, and for many requests a cheaper model gives an answer that is just as good. The practical question is which model to use for a given request so that quality stays high and cost stays low. This matters to any team running LLMs at scale, where token cost adds up quickly and the price gap between models can be large.

LLMRouterBench fits this problem well. It contains 23,945 prompts answered across multiple models, which expands to 391,645 individual records. Each record stores the prompt, the prompt and completion token counts, the dollar cost, and a quality score for that model on that prompt. The dataset is public and comes as precomputed results, so it is real measured data rather than something I generate, and it needs reshaping more than heavy cleaning.

I will frame two tasks that use two different types of algorithms studied in this course. The first is classification: for each request, predict which model is the cost-optimal choice, where I build the label from the recorded score and cost (the cheapest model that clears a quality threshold). The second is regression: predict a request's cost so the router can weigh the tradeoff

before it commits. For the experimental comparison I will compare algorithms within each task, for example logistic regression against a random forest for the routing decision, and I will compare the learned router against simple baselines such as always pick the cheapest model and always pick the strongest. If it adds value, I will use K-means clustering to group prompts by economic profile.

The build will be a scikit-learn pipeline in a documented Jupyter notebook that loads the data, runs exploratory data analysis, engineers a small set of features, trains and tunes the models, and evaluates them. The code will live in a GitHub repository with a README and a ruff linting workflow, following the PEP 8 style guide. Outputs will include cost-versus-quality curves, a confusion matrix for the routing decision, and feature-importance plots.

Related Course Topics

This project uses supervised classification and regression, the machine learning lifecycle from data preparation through evaluation, feature selection and feature importance, model evaluation, and hyperparameter tuning. It also touches unsupervised clustering as an optional step. These map to the core machine learning material in Poole and Mackworth (2023, Chapter 7).

Expected System Behaviors

A good router should send easy requests to cheaper models and hard requests to stronger ones, cutting total cost while keeping quality close to the strongest model. Published routing work reports cost reductions near 85 percent on some benchmarks while keeping most of the strong model's quality, so a meaningful saving at high quality is a reasonable target. I expect prompt length and task type to be the features that most influence the routing decision, and I

expect the cheapest-always and strongest-always baselines to sit at the two ends of the cost-quality tradeoff, with the learned router landing in the useful middle.

Issues of Focus

Four issues will get the most attention. First, defining the routing label honestly, since it depends on a chosen quality threshold and on how I weigh cost against quality; I will state that choice and test how sensitive the results are to it. Second, measuring success as cost saved at a given quality level rather than accuracy alone, because raw accuracy does not capture the money saved. Third, avoiding data leakage, since the same prompt appears across models and must not sit on both sides of the train and test split. Fourth, keeping the prompt features light, using signals like prompt length and task category rather than heavy text embeddings, so the work stays within the light-preprocessing expectation for this project.

AI tool disclosure: I used generative AI (Claude) as a thinking partner to aid my understanding of the course material and to help organize this proposal, including pressure-testing the dataset choice and outlining the two modeling tasks. I have reviewed, modified, and verified the AI-assisted text against the cited sources, and the methodology, claims, and conclusions here reflect my own understanding. The cost-aware routing framing and the accountant-for-tokens problem statement are my own.

References

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